A multisynaptic spiking neuron for simultaneously encoding spatiotemporal dynamics

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Introduction

Motivation

- Real-world sensory data (vision, audio, biological signals) are inherently spatiotemporal
- Biological brains process such signals rapidly and efficiently
- Spiking Neural Networks (SNNs) are promising due to their temporal dynamics and energy efficiency

Research Gap

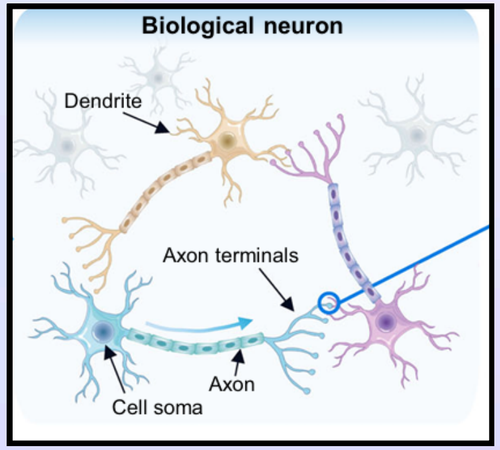
- Despite theoretical advantages, SNNs underperform in practical spatiotemporal tasks
- Existing improvements focus mainly on training algorithms and learning rules
- The role of neuron-level information representation remains underexplored

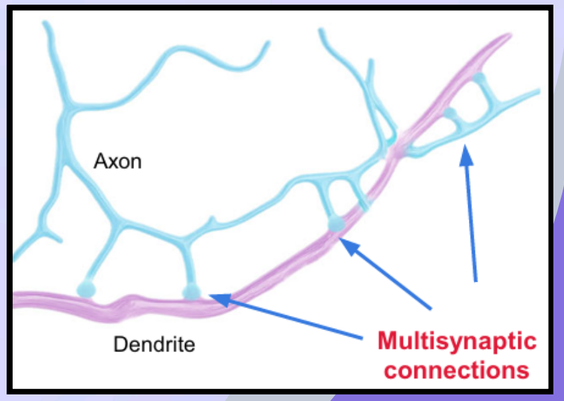
Key Insight of This Paper

- The performance gap arises from information loss at the neuron level
- Single-synapse spiking neurons are insufficient for rich spatiotemporal encoding

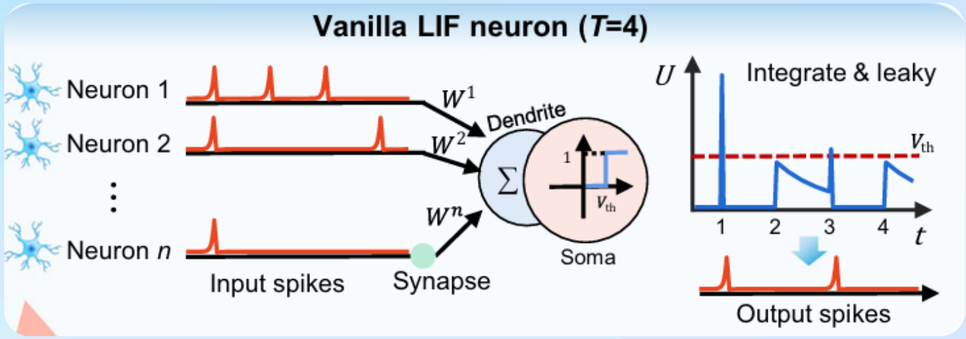
Contribution

- Proposes a Multi-Synaptic Firing (MSF) neuron
- Enables simultaneous encoding of spatial intensity and temporal dynamics
- Advances SNNs toward real-world spatiotemporal applications





SNN (Spiking Neural Network)



$$\tau \frac{du}{dt} = -u + I,$$

$$s(t) = \Theta(u(t) - V_{th})$$

- Spiking Neural Networks process information using discrete spikes over time, not continuous values
- Information is encoded using spike timing (when spikes occur) and spike rate (how often they occur)
- The most common spiking neuron is the Leaky Integrate–and–Fire (LIF) neuron
- Incoming spikes are weighted, summed, and integrated, while the membrane potential leaks over time

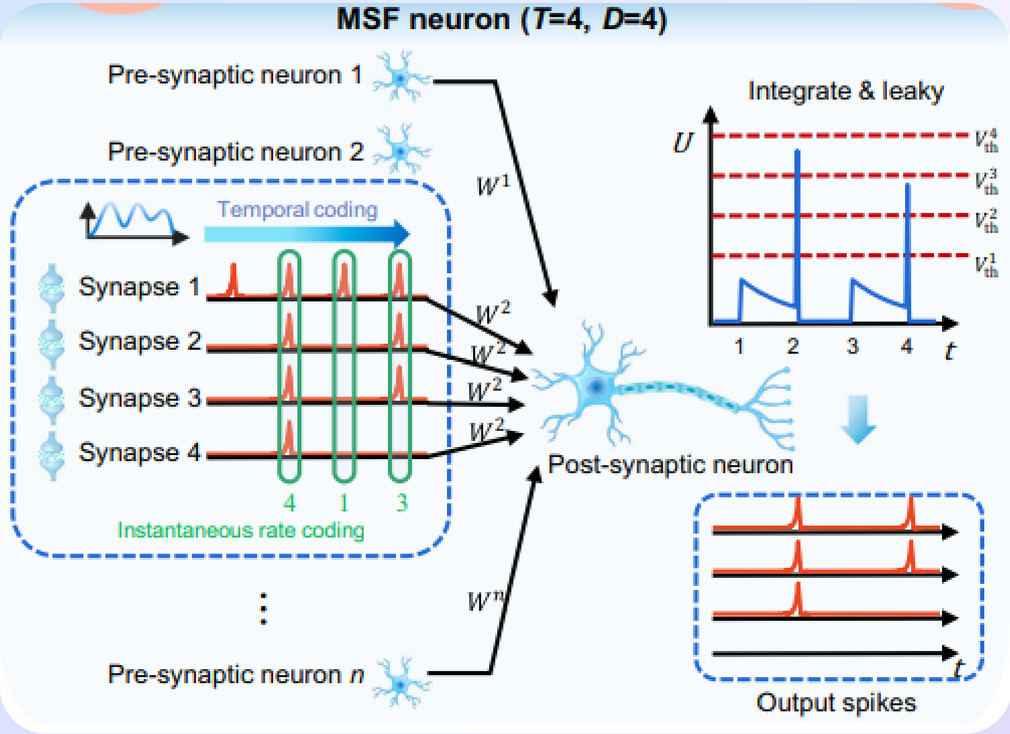
- A spike is generated when the membrane potential crosses a fixed threshold, and then the neuron resets
- Output spikes are binary, with at most one spike per timestep
- LIF neurons can encode either spike timing or firing rate, but not both simultaneously
- Strong and weak inputs may produce similar spike outputs, causing information loss

MSF MODEL

- To overcome LIF's limitation, the MSF neuron introduces multisynaptic connections between neurons.
- Each pre-synaptic neuron connects to a post-synaptic neuron through multiple synapses, rather than a single channel.
- All synapses share the same membrane potential, but each synapse is associated with a different firing threshold.

- Strong input ⇒ multiple thresholds crossed & multiple spikes at same timestep
- Weak input ⇒ few or no spikes

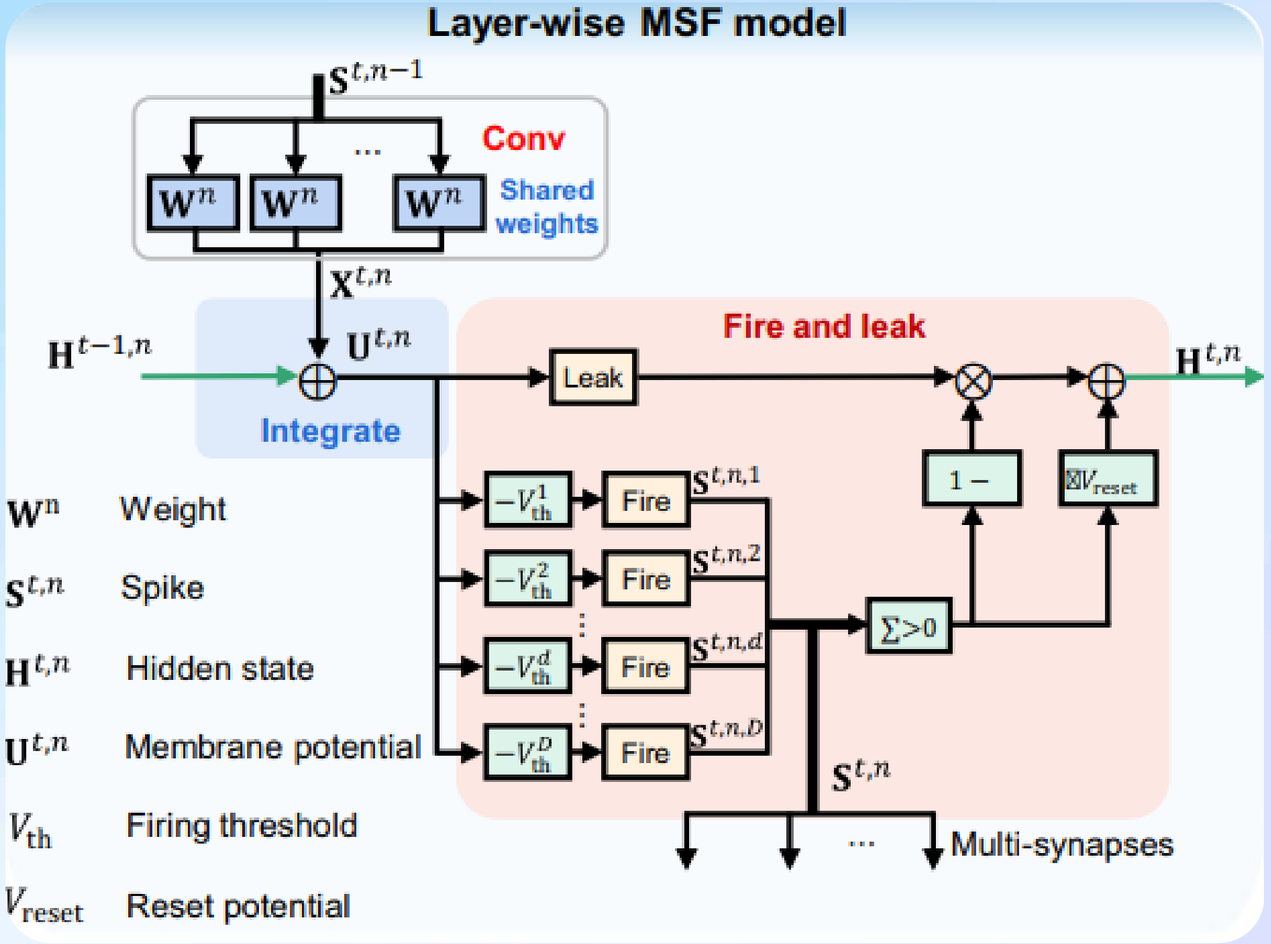
- In this way, the number of spikes generated at a timestep encodes stimulus intensity, while the exact timing of spikes preserves temporal information.
- Unlike LIF neurons, multiple spikes can occur in a single timestep, and the neuron resets only when at least one threshold is crossed.
- As a result, the encoding capacity of the neuron scales with the number of synapses D.



- MSF neuron model developed from numerically discretized LIF is formulated as follows:

$$
\begin{cases}
u_i^{t,n} = u_{\text{reset}} \cdot \text{sign}\left(\sum_{d=1}^{D} s_i^{t-1,n,d}\right) + \beta u_i^{t-1,n}\left(1 - \text{sign}\left(\sum_{d=1}^{D} s_i^{t-1,n,d}\right)\right) \\
\qquad + \sum_j \sum_{d=1}^{D} w_{ij}^{n,d} s_j^{t,n-1,d} \\
s_i^{t,n,d} = \Theta\left(u_i^{t,n} - V_{th}^{d}\right) = \begin{cases} 1, & \text{if } u_i^{t,n} \geq V_{th}^{d}, \\ 0, & \text{otherwise} \end{cases} \quad d = 1, 2, \ldots, D
\end{cases}
$$

Layer-wise MSF Model



Spatial feature extraction:

$$X^{t,n} = \sum_{d=1}^{D} W^{n,d} S^{t,n-1,d}$$

Membrane integration:

$$U^{t,n} = H^{t-1,n} + X^{t,n}$$

Spiking:

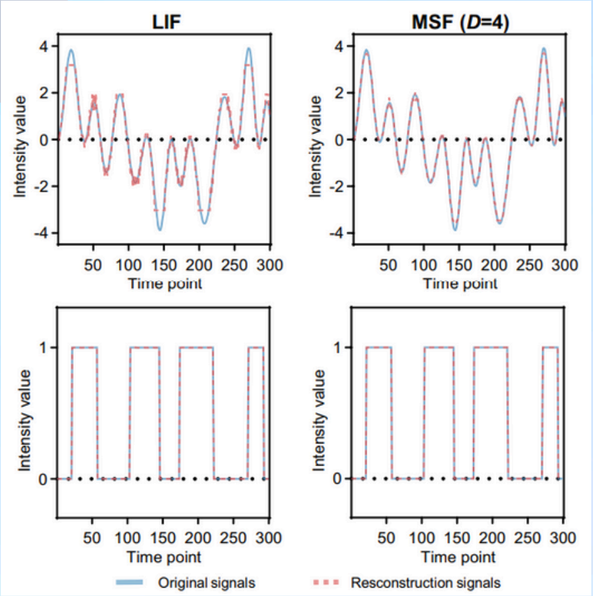
$$S^{t,n,d} = \Theta(U^{t,n} - V_{th}^d)$$

Hidden state update:

$$H^{t,n} = V_{reset} \cdot \mathrm{sign}\left(\sum_d S^{t,n,d}\right) + \beta U^{t,n}\left(1 - \mathrm{sign}\left(\sum_d S^{t,n,d}\right)\right)$$

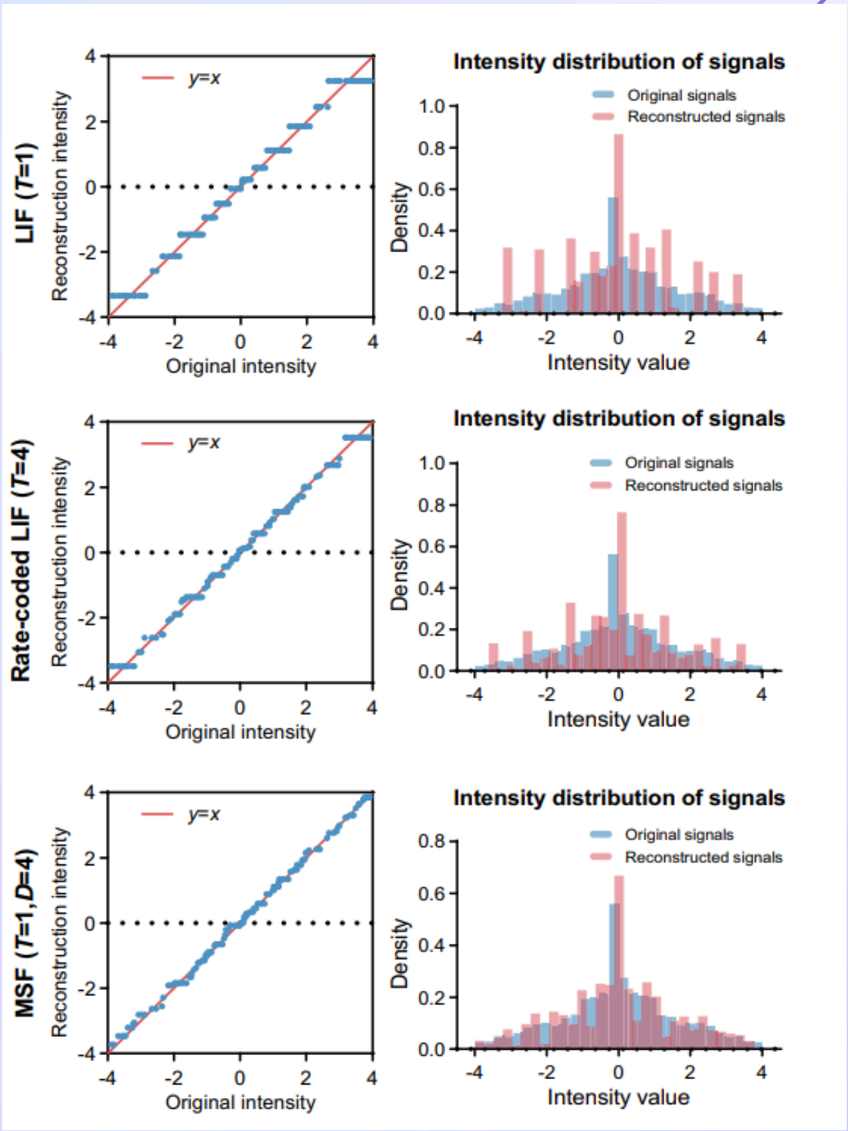
- Shared weights across synapses: $W^{n,1}=\ldots=W^{n,D}$
- No increase in parameter count
- MSF Neurons Generalize LIF and ReLU
- Special cases:
  - D = 1 → MSF reduces to LIF
  - T = 1, D → ∞ → MSF approximates ReLU
- It bridges Spiking Neural Networks (SNNs) & Artificial Neural Networks (ANNs)

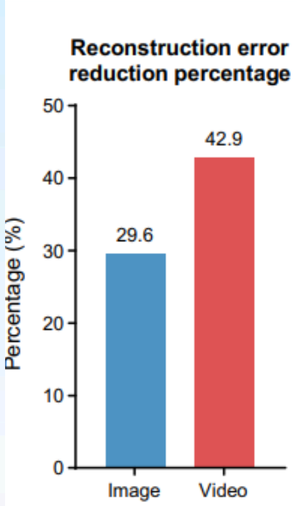
Signal reconstruction using spiking autoencoder

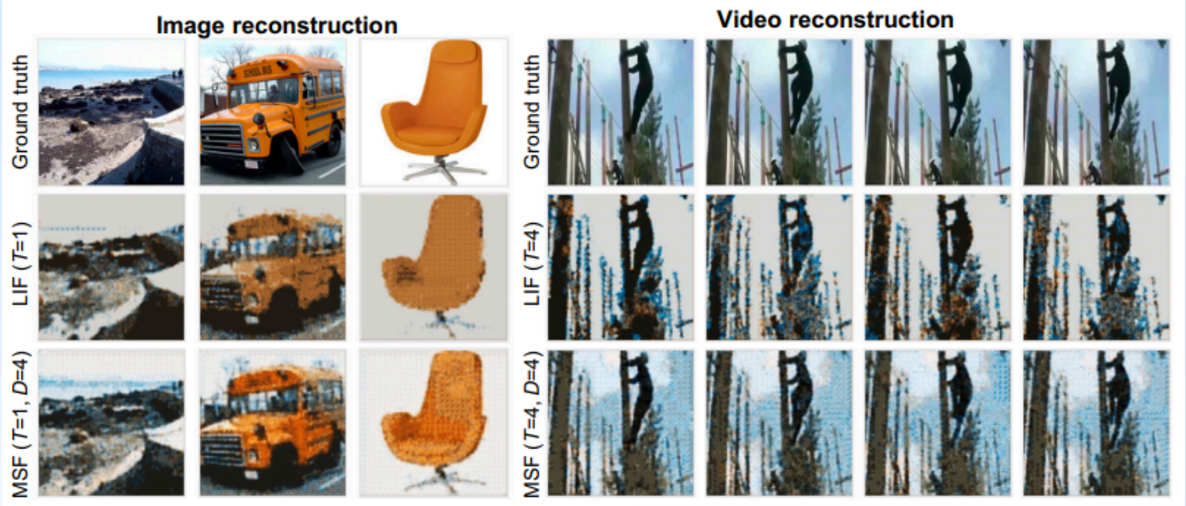


Reconstructed sine and square wave temporal signals

Reconstructed temporally compressed versions of the sinusoidal signals, representing static signals of different intensities.







Surrogate Gradient

From the chain rule of backpropagation, we have

$$\frac{\partial \mathcal{E}}{\partial \mathbf{S}^l} = \frac{\partial \mathcal{E}}{\partial \mathbf{S}^L}\frac{\partial \mathbf{S}^L}{\partial \mathbf{S}^l} = \frac{\partial \mathcal{E}}{\partial \mathbf{S}^L}\cdot (f')^{L-l}\cdot \prod_{i=l}^{L-1}\frac{\partial F(\mathbf{S}^i, \mathbf{W}^{i+1})}{\partial \mathbf{S}^i},$$

In the backpropagation process, the gradient is discontinuous at multiple locations. The surrogate gradient of MSF consists of multiple surrogate functions, which can be described as:

$$\frac{\partial \mathbf{S}^{t,n}}{\partial \mathbf{U}^{t,n}} = \sum_{d=1}^{D}\Theta'(\mathbf{U}^{t,n} - V_{\text{th}}^{d}) \approx \sum_{d=1}^{D}\sigma(\mathbf{U}^{t,n} - V_{\text{th}}^{d}). \tag{9}$$

Therefore, the setting of thresholds and the choice of surrogate functions directly determine the gradients of the activation functions during the backpropagation process. To proportionally measure the magnitude of membrane potential, it is generally necessary to have equal intervals between thresholds when selecting multiple threshold values. By setting an initial threshold V_{th} and a threshold interval $h > 0$, the surrogate gradient $f'(\mathbf{U}^{t,n})$ of MSF can be described as follows:

$$f'(\mathbf{U}^{t,n}) = \sum_{d=1}^{D}\sigma(\mathbf{U}^{t,n} - V_{\text{th}}^{d}) = \sum_{d=0}^{D-1}\sigma(\mathbf{U}^{t,n} - V_{\text{th}} - dh). \tag{10}$$

To simplify the expression, we can shift the summation index to make the summation symmetric about zero. Define a new index $d' = d - \frac{D-1}{2}$, which shifts the range of summation to:

$$f'(\mathbf{U}^{t,n}) = \sum_{d'=-\frac{D-1}{2}}^{\frac{D-1}{2}}\sigma(\mathbf{U}^{t,n} - V_{\text{th}} - d'h). \tag{11}$$

When $D \to \infty$, the summation can be approximated by a continuous integral. By approximating the summation over discrete d' with an integral over the continuous variable $x = d'h$, we obtain:

$$f'(\mathbf{U}^{t,n}) \approx \frac{1}{h}\int_{-\infty}^{\infty}\sigma(\mathbf{U}^{t,n} - V_{\text{th}} - x)\,dx. \tag{12}$$

Let $y = \mathbf{U}^{t,n} - V_{\text{th}} - x$, which gives $dy = -dx$. Substituting this into the integral, we get:

$$f'(\mathbf{U}^{t,n}) \approx \frac{1}{h}\int_{\infty}^{-\infty}\sigma(y)(-dy) \approx \frac{1}{h}\int_{-\infty}^{\infty}\sigma(y)\,dy. \tag{13}$$

Given that the surrogation gradient function $\sigma(x)$ satisfies $\int_{-\infty}^{\infty}\sigma(x)\,dx = 1$, we can obtain:

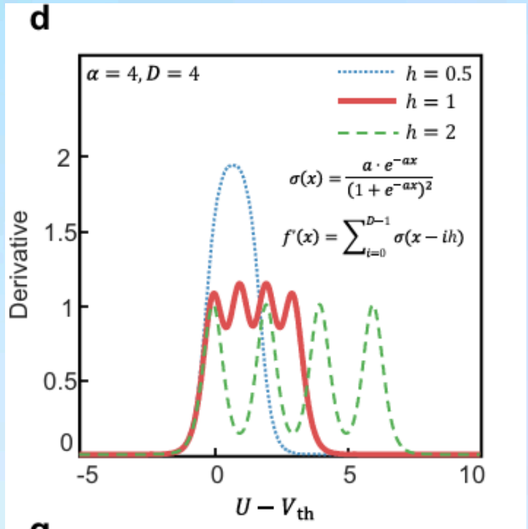
$$f'(\mathbf{U}^{t,n}) \approx \frac{1}{h}. \tag{14}$$

- The spike function in spiking neurons is non-differentiable, causing discontinuous gradients, so surrogate gradients are used to approximate the gradient near firing thresholds during backpropagation
- In MSF neurons, having multiple thresholds results in multiple surrogate gradient contributions, and the total gradient is obtained by summing these individual surrogate gradients

- The choice of thresholds and surrogate gradient function directly controls gradient behavior during backpropagation
- For MSF neurons, multiple thresholds should be equally spaced to correctly measure the membrane potential
- Using an initial threshold Vth and a threshold interval h, the MSF surrogate gradient is defined

- For stable training, the surrogate gradient should stay close to 1 to avoid vanishing or exploding gradients
- Setting the threshold interval h=1 gives the most stable gradient behavior, regardless of surrogate function type
- The steepness parameter α controls gradient oscillation and should be adjusted to improve training stability

Surrogate Gradient Behavior in MSF Neurons (Effect of Threshold Spacing and Curve Steepness)



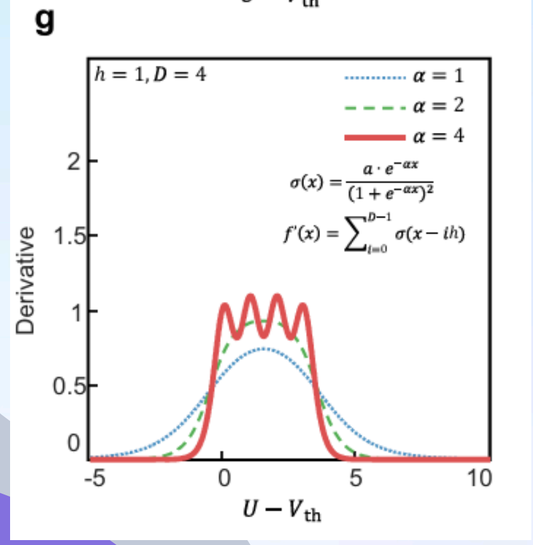


Figure d: (Effect of Threshold Spacing (h))

- MSF neurons use multiple thresholds, each contributing a surrogate gradient
- When h is small:
  - Thresholds are close
  - Gradients overlap strongly
  - Total gradient becomes too large → exploding gradients
- When h is large:
  - Thresholds are far apart
  - Gradients overlap less
  - Total gradient becomes too small → vanishing gradients
- When h is approximately 1, it gives a balanced gradient close to 1

Figure g: (Effect of Curve Steepness (α))

- α controls how sharp or smooth each surrogate gradient is
- Small α:
  - Very smooth gradients
  - Weaker learning signal
- Large α:
  - Very sharp gradients
  - High oscillations and instability
- Moderate α gives smooth and stable learning

Conclusion :

- Threshold spacing (h) controls gradient magnitude
- Curve steepness (α) controls gradient smoothness
- Choosing h ≈ 1 and moderate α keeps gradients stable
- This enables deep and accurate MSF-based spiking neural networks

Thank You